ToricGT: Compact Graph-Token Reasoning with Tropical Ring Attention, Toric Diagnostics, and Soft-MoE Routing

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Abstract

We present ToricGT, a compact graph-to-graph Transformer family for algebraic, mathematical, morphological, and parameter-constrained language-model reasoning. The model tokenizes nodes and edges as a shared sequence, uses exact blockwise ring evaluation for tropical max-plus attention, adds finite noncommutative-torus phase channels, and replaces upper feed-forward blocks by permutation-equivariant Soft-MoE routing. The paper contributes a finite theory and an implementation-facing training contract. We prove graph-token equivariance, exactness of tropical ring attention, stability of active faces under perturbation, and simulation of bounded semiring dynamic programs. Motivated by recent toric geometry of ReLU networks, we further interpret tropical head decisions as active faces of Newton polytopes and normal fans, giving compact diagnostics for margins, bends, and future toric auxiliary losses. A separate Parameter-Golf track distills the graph model into a byte-limited micro language model with causal Soft-MoE and quantization-aware export. The result is a small, auditable model design whose reasoning behavior is inspected through equivariance errors, active faces, toric moments, expert utilization, and GFlowNet diversity rather than scalar accuracy alone.

1 Introduction

Many reasoning tasks are naturally graph-to-graph transformations: proof steps depend on earlier equations, graph algorithms update node and edge states, Hebrew morphology links surface forms to roots and templates, and algebraic words reduce through local rewrite relations. ToricGT targets this regime with three design constraints. First, relabeling a graph should relabel node and edge outputs, not change the computation. Second, dynamic-programming-like reasoning should be visible through max-plus active faces and margins. Third, compactness matters: the same ideas should support a small research model and a Parameter-Golf-style micro language model under a strict artifact budget.

The main novelty is the integration of algebraic structure with trainable graph-token Transformers. Tropical attention implements max-plus gates and exposes provenance. Ring evaluation changes the schedule but not the function. Finite Weyl pairs encode projective toric phase memory. Soft-MoE gives typed differentiable routing without hard token dropping. Finally, toric geometry supplies a new diagnostic layer: active candidates in tropical heads form exposed faces of Newton polytopes, and changes across cell walls behave like bend or Cartier-divisor intersection measurements.

2 Model

A ToricGT instance is

$$F(G) = D \circ \Phi_L \circ \cdots \circ \Phi_1 \circ T(G), \tag{1}$$

where T maps a typed attributed graph to node and edge tokens, Φ_ℓ is a shared token-equivariant block, and D is a shared node/edge/graph decoder. A graph is

$$G = (V, E, s, t, \tau_V, \tau_E, x, e, m_V, m_E), \tag{2}$$

with source and target maps for edge tokens, finite node and edge types, bounded attributes, and padding masks.

Tropical ring attention. For score matrix S and values V, a tropical head computes

$$Y_{ic} = \max_j (S_{ij} + V_{jc}), \qquad A_{ic} = \arg\max_j (S_{ij} + V_{jc}). \tag{3}$$

Partitioning keys into blocks $B_1, \ldots, B_R$ gives

$$\max_j (S_{ij} + V_{jc}) = \max_r \max_{j \in B_r} (S_{ij} + V_{jc}), \tag{4}$$

so ring evaluation is exactly equal to full tropical attention in exact arithmetic.

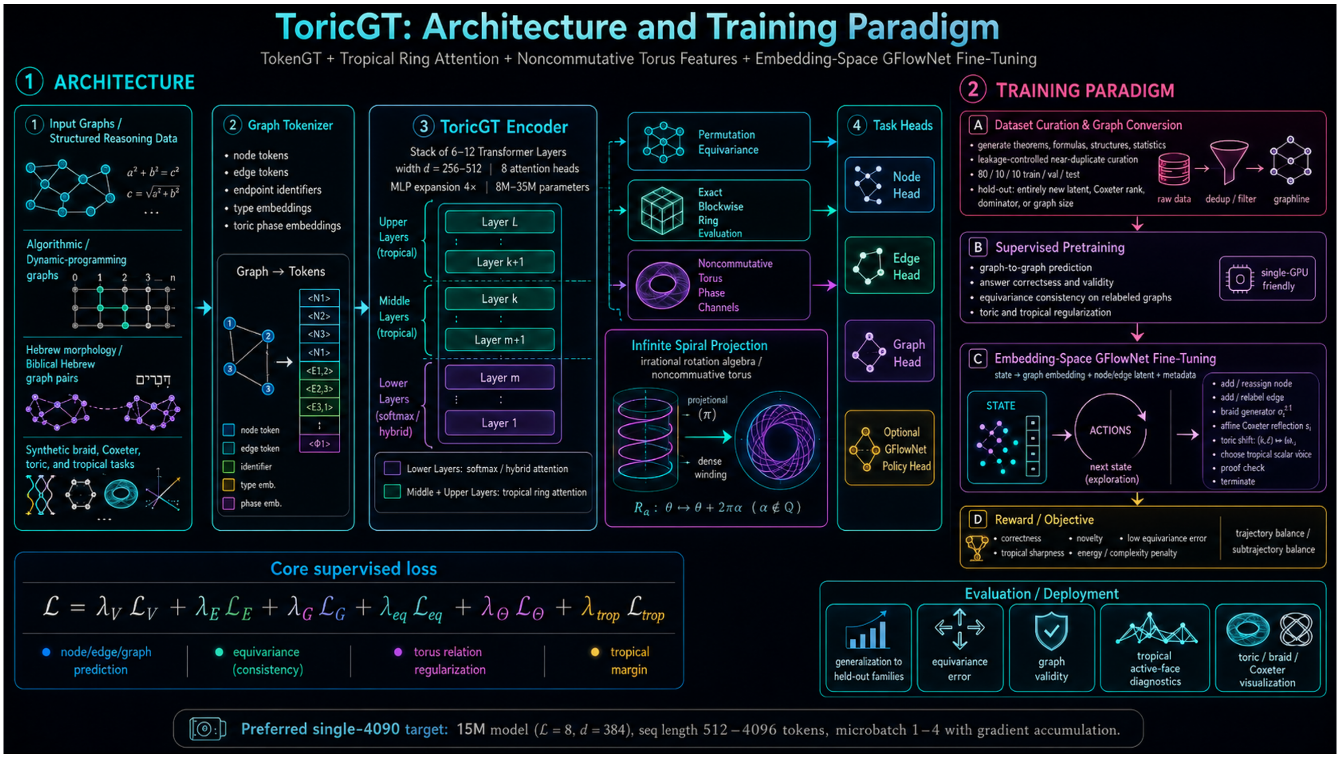

Figure 1: ToricGT combines graph tokenization, tropical ring attention, noncommutative-torus phase channels, Soft-MoE routing, and embedding-space GFlowNet fine-tuning.

Soft-MoE routing. For hidden states $X \in \mathbb{R}^{B \times L \times d}$, learned slot vectors $\Phi \in \mathbb{R}^{d \times E \times S}$, and validity mask M, define

$$A_{bies} = g \langle x_{bi} / \|x_{bi}\|, \phi_{es} / \|\phi_{es}\| \rangle, \tag{5}$$

$$D_{bies} = \frac{\exp A_{bies} M_{bi}}{\sum_j \exp A_{bjes} M_{bj} + \epsilon}, \tag{6}$$

$$C_{bies} = \frac{\exp A_{bies}}{\sum_{e',s'} \exp A_{bie's'} + \epsilon}, \tag{7}$$

$$Z_{bes} = \sum_i D_{bies} X_{bi}, \qquad Y_{bi} = \sum_{e,s} C_{bies} f_e(Z_{bes}). \tag{8}$$

We use $E \geq 4$ experts by default in the graph model. The Parameter-Golf micro-LM uses a prefix-causal variant in which slot accumulators at position t contain only strict-prefix tokens.

3 Theory

Theorem 1 (Graph equivariance)**.** *If tokenization, masks, attention projections, tropical heads, Soft-MoE routing, residuals, normalization, and decoders are shared over token rows and use only equivariant incidence/type information, then ToricGT is graph-to-graph equivariant.*

Proof. Tokenization maps relabeled graphs to row-permuted token tables. Linear projections commute with row permutations. Softmax attention, tropical attention, and Soft-MoE dispatch/combination commute with the same row permutation because their normalizers sum over either all valid tokens or slots for a fixed token. Row-wise residual, normalization, and decoder maps commute with permutations. Composition preserves equivariance. □

Lemma 2 (Active-face stability)**.** *If* $\left\|S - \widehat{S}\right\|_\infty \leq \epsilon_S$ *and* $\left\|V - \widehat{V}\right\|_\infty \leq \epsilon_V$*, then tropical outputs obey*

$$\left\|Y - \widehat{Y}\right\|_\infty \leq \epsilon_S + \epsilon_V. \tag{9}$$

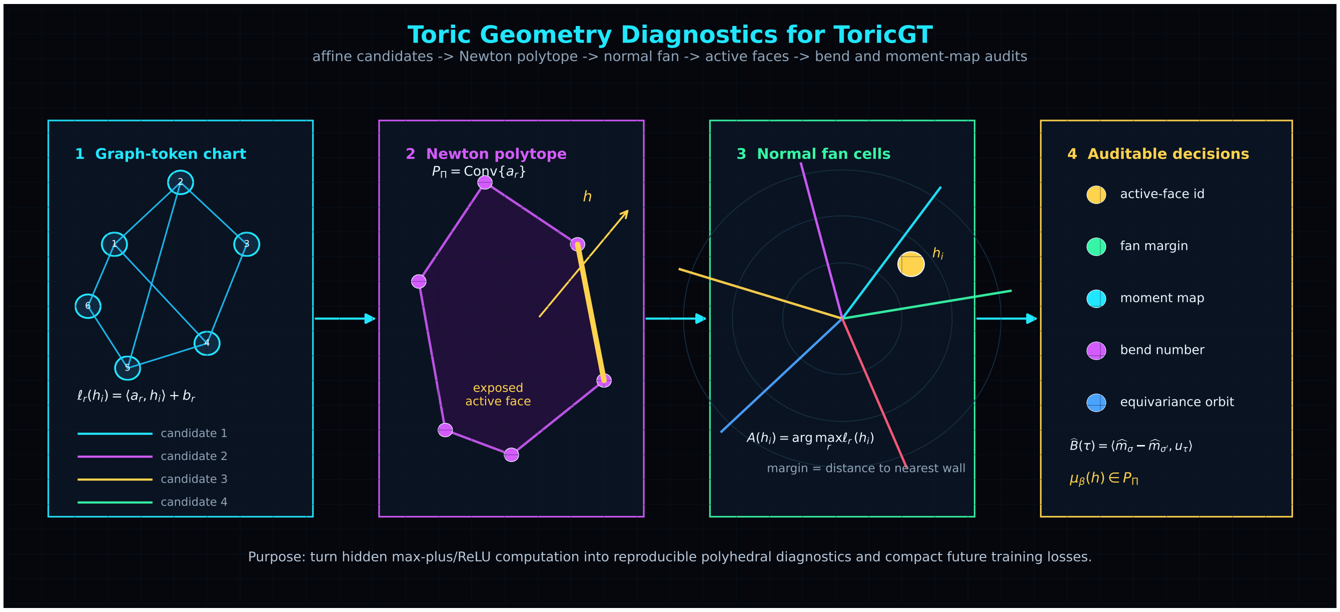

Figure 2: Toric diagnostics turn tropical candidate scores into Newton polytopes, normal-fan cells, active-face margins, and bend measurements.

If the top-two margin is larger than $2(\epsilon_S + \epsilon_V)$, the active key is unchanged.

Proof. Each candidate score changes by at most $\epsilon_S + \epsilon_V$, and the maximum is 1-Lipschitz in ℓ_∞. The margin condition keeps the original maximizer strictly above all competitors after perturbation. □

Toric diagnostic theorem. Let a tropical probe be $\psi(h) = \max_r(\langle a_r, h\rangle + b_r)$ with rational slopes a_r. Its Newton polytope is $P = \mathrm{Conv}(a_1, \ldots, a_R)$. The active set

$$A(h) = \arg\max_r(\langle a_r, h\rangle + b_r) \tag{10}$$

is the exposed face of the lifted polytope $\mathrm{Conv}\{(a_r, b_r)\}_r$ under the linear functional $(a, b) \mapsto \langle a, h\rangle + b$. Hence tropical active-face diagnostics are cells of a normal fan. This imports the ReLU fan / Cartier-divisor viewpoint of Fu [1] into ToricGT tropical subcomputations, complementing earlier tropical rational interpretations of ReLU networks [2].

4 Training And Evaluation

The supervised objective combines node, edge, and graph losses with equivariance, torus, tropical-margin, Hebrew morphology, Soft-MoE, cache, and distillation terms:

$$\begin{aligned} \mathcal{L} = {} & \lambda_V \mathcal{L}_V + \lambda_E \mathcal{L}_E + \lambda_G \mathcal{L}_G + \lambda_{\mathrm{eq}} \mathcal{L}_{\mathrm{eq}} + \lambda_\Theta \mathcal{L}_\Theta + \lambda_{\mathrm{trop}} \mathcal{L}_{\mathrm{margin}} \\ & + \lambda_H \mathcal{L}_H + \lambda_{\mathrm{moe}} \mathcal{L}_{\mathrm{moe}} + \lambda_{\mathrm{cache}} \mathcal{L}_{\mathrm{PQ}} + \lambda_{\mathrm{KD}} \mathcal{L}_{\mathrm{distill}}. \end{aligned} \tag{11}$$

Embedding-space GFlowNet fine-tuning samples reasoning graphs with probability proportional to reward using trajectory balance. The reward combines correctness, novelty, low equivariance error, tropical margin, low toric commutator error, and complexity penalties.

The data mixture is graph-first: math reasoning, graph-of-thought traces, algorithmic tasks, synthetic rotation/tropical tasks, Hebrew morphology graphs, and Jewish Hebrew text sources are normalized into Parquet records with stable content hashes and graph JSON. Splits are clustered by near-duplicate text, graph sketches, algebraic normal forms, root families, and source references.

Metric	Failure mode exposed
Equivariance error	absolute-position leakage and edge-order bugs
Active-face margin	unstable dynamic-programming decisions
Toric bend consistency	incoherent polyhedral reasoning changes
Soft-MoE expert mass	expert collapse or over-routing
GFlowNet diversity	mode collapse and reward hacking
Parameter-Golf BPB/bytes	compactness and scoring compliance

5 Parameter-Golf Track

The contest-facing model is not the full graph encoder. It is a micro language model distilled from graph-reasoning traces with tied embeddings, recurrent block reuse, grouped-query attention, prefix-causal Soft-MoE adapters, and quantization-aware export. The artifact budget is

$$B_{\text{art}} = B_{\text{code}} + B_{\text{weights}} + B_{\text{tokenizer}} + B_{\text{metadata}} \leq 16{,}000{,}000. \tag{12}$$

BPB is computed prequentially: predict a normalized next-symbol distribution, score the observed bytes, then update any legal online state. This score-before-update rule is mandatory for causal Soft-MoE and test-time adaptation.

6 Next Iteration

The next ToricGT iteration should use toric geometry as a training signal. First, freeze the trained checkpoint and build empirical toric shadows: occupied fan cells, fitted slopes, active-face margins, and bend magnitudes. Second, add synthetic toric tasks: Newton-polytope active-face prediction, Cartier-divisor bend prediction, toric ideal relation checks, and ReLU fan realization tests. Third, attach low-rank quantized toric probes and optimize

$$\mathcal{L}_{\text{toric}} = \mathcal{L}_{\text{base}} + \lambda_{\text{fan}}\mathcal{L}_{\text{fan}} + \lambda_{\text{bend}}\mathcal{L}_{\text{bend}} + \lambda_{\text{binom}}\mathcal{L}_{\text{binom}} + \lambda_{\text{moment}} \left\|\widehat{\mu} - \mu^{\star}\right\|_2^2. \tag{13}$$

Finally, make inference-time scaling geometric: continuous GFlowNets can move inside normal cones, along walls, and across moment polytopes while discrete actions edit graph-of-thought states. Continuous GFlowNet theory supplies the density-correct trajectory-balance objective for these hybrid paths [7]. This keeps the architecture small while giving search a structured latent space.

7 Conclusion

ToricGT is a finite, auditable model family. Its exact claims are graph-token equivariance, ring-schedule exactness, max-plus semiring simulation on bounded domains, projective toric phase consistency, causal Soft-MoE for language scoring, and perturbation bounds for compressed caches. Its practical claim is that reasoning models should expose the geometry of their decisions. Tropical active faces, Newton polytopes, toric bends, expert utilization, and GFlowNet reward landscapes make small-model reasoning easier to test, compress, and improve.

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